



PREDator

Prediction & Forecast Engine

USER GUIDE

Everything you need to run your first forecast.

REQUIREMENTS

Before launching PREDator make sure you have the following ready:

- Your time-series dataset saved as a .csv file.
- A folder where PREDator will save all outputs.
- The name of the column you want to forecast (target variable).

LAUNCHING THE APP

Double-click PREDator.app. On first launch, config.yaml is created automatically next to the app. Make sure config.yaml is present in the same folder as PREDator.app before the first run.

STEP-BY-STEP WORKFLOW

1. Select dataset — click the button and navigate to your .csv file.
2. Select output folder — choose where all results will be saved.
3. Set parameters:
 - forecast_horizon — how many future steps to predict (H).
 - look_back — length of the input rolling window (L) (RNN only).
 - n_epochs — number of training iterations (RNN only).
 - target_var — exact column name in your CSV to forecast.
4. Choose model — RNN (neural network) or GPR (Gaussian Process Regression).
 - If GPR is selected, only target_var and look_back are required. The other fields are disabled.
5. Click RUN PIPELINE. The console streams live training progress.
6. When complete, RMSE and NRMSE are printed. All results are saved automatically.

OUTPUT FOLDER STRUCTURE

After each run, PREDator generates the following inside your chosen output folder:

- forecasts/model/model.h5 — serialized model weights.
- forecasts/csv/training_loss/ — loss value per epoch (.csv).
- forecasts/csv/predictions/ — forecast values per epoch (.csv).
- forecasts/grafici/ — forecast plots (.png).

- forecasts/rmse.txt — final RMSE and NRMSE values.
- gpr/ — GPR model, forecast CSV and plot (GPR mode only).
- logs/ — execution log file.

TIPS

- If a run fails, read the console — the error message shows exactly what went wrong.
- You can change parameters and re-run without restarting the app.
- config.yaml next to the app is updated automatically at each run and can also be edited manually.